Alternating Current (Supplementary Problems)

1. What is the reactance of a 1200-pF capacitor at 400 Hz?

2. A capacitor has a reactance of 450 Ω at 1200 Hz. What is its capacitance?

3. The AC voltage across a capacitor with C = 1.5 μF has an amplitude of V_{max} = 22 V, and the current through the capacitor has an amplitude of 2.5 mA. What is the frequency?

4. Two capacitors are connected to an AC source as shown in Figure P22.25; assume V_{AC} = Vmax sin(2πft), with V_{max} = 18 V and a frequency f = 2000 Hz. (a) What is the amplitude (peak value) of the current through the capacitors? (b) What is the rms current?

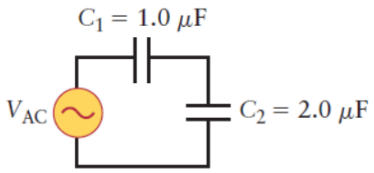


Figure P22.25

5. Two capacitors with $C_1 = 4.5\ \mu F$ and $C_2 = 1.9\ \mu F$ are connected in series. What is the reactance of the equivalent capacitance at 350 Hz?

6. Two capacitors with $C_1 = 450\ pF$ and $C_2 = 820\ pF$ are connected in parallel. What is the reactance of the equivalent capacitance at 30 kHz?

7. You are given two inductors with inductances of 40 mH and 80 mH. If the frequency is 5000 Hz, which inductor has the larger reactance?

8. An inductor has a reactance of 5500 Ω at a frequency of 2500 Hz. What is the value of L?

9. Find the reactance of an inductor with L = 15 mH at 22 kHz. If the frequency is increased to 66 kHz, by what factor does the reactance change?

10. Consider the circuit in Figure P22.36. (a) If the current through the inductor (L = 25 mH) at 400 Hz has an amplitude of 55 mA, what is the amplitude of the AC voltage? (b) What is the rms value of the AC voltage across the inductor?

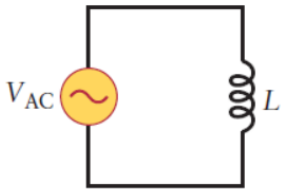


Figure P22.36

11. The voltage source in Figure P22.38 has an amplitude of 5.0 V and a frequency of 75 Hz. What is the amplitude of the current through the inductors?

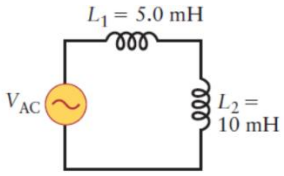


Figure P22.38

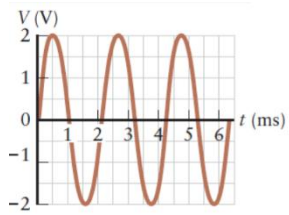


Figure P22.39

12. Figure P22.39 shows the voltage across an inductor with L = 20 mH. (a) At what time is the energy stored in the inductor a maximum? What is the energy at that moment? (b) At what time is the energy stored in the inductor a minimum? What is the energy at that moment?

13. Consider an inductor with L = 50 mH and a resistor with R = 20 kΩ. At what frequency is the reactance of the inductor equal to the resistance of this resistor?

14. A sinusoidal, б0.0-Hz, ac voltage is read to be 120 V by an ordinary voltmeter. (*a*) What is the maximum value the voltage takes on during a cycle? (*b*) What is the equation for the voltage?

15. A voltage v= б0.0 V sin 120πt is applied across a 20.0-Ω resistor. What will an ac ammeter in series with the resistor read?

16. A 120-V ac voltage source is connected across a 2.0-μF capacitor. Find the current to the capacitor if the frequency of the source is (*a*) 60 Hz and (*b*) 60 kHz. (*c*) What is the power loss in the capacitor?

17. A 120-V ac voltage source is connected across a pure 0.700-H inductor. Find the current through the inductor if the frequency of the source is (*a*) 60.0 Hz and (*b*) 60.0 kHz. (*c*) What is the power loss in the inductor?

18. A coil having inductance 0.14 H and resistance of 12 Λ is connected across a 110-V, 25-Hz line. Compute (*a*) the current in the coil, (*b*) the phase angle between the current and the supply voltage, (*c*) the power factor, and (*d*) the power loss in the coil.

19. A capacitor is in series with a resistance of 30 Ω and is connected to a 220-V ac line. The reactance of the capacitor is 40 Ω. Determine (*a*) the current in the circuit, (*b*) the phase angle between the current and the supply voltage, and (*c*) the power loss in the circuit.

20. A series circuit consisting of a 100-Ω noninductive resistor, a coil with a 0.10-H inductance and negligible resistance, and a 20-μF capacitor is connected across a 110-V, 60-Hz power source. Find (*a*) the current, (*b*) the power loss, (*c*) the phase angle between the current and the source voltage, and (*d*) the voltmeter readings across the three elements.

21. A 5.00-Ω resistance is in a series circuit with a 0.200-H pure inductance and a 40.0-nF pure capacitance. The combination is placed across a 30.0-V, 1780-Hz power supply. Find (*a*) the current in the circuit, (*b*) the phase angle between source voltage and current, (*c*) the power loss in the circuit, and (*d*) the voltmeter reading across each element of the circuit

22. As shown in Fig. 35-3, a series circuit connected across a 200-V, 60-Hz line consists of a capacitor of capacitive reactance 30 Ω, a noninductive resistor of 44 Ω, and a coil of inductive reactance 90 Λ and resistance 36 Ω. Determine (*a*) the current in the circuit, (*b*) the potential difference across each element, (*c*) the power factor of the circuit, and (*d*) the power absorbed by the circuit.

23. Calculate the resonant frequency of a circuit of negligible resistance containing an inductance of 40.0 mH and a capacitance of 600 pF.

24. A step-up transformer is used on a 120-V line to furnish 1800 V. The primary has 100 turns. How many turns are on the secondary?

25. A transformer used on a 120-V line delivers 2.0 A at 900 V. What current is drawn from the line? Assume 100 percent efficiency.

26. A step-down transformer operates on a 2.5-kV line and supplies a load with 80 A. The ratio of the primary winding to the secondary winding is 20:1. Assuming 100 percent efficiency, deter- mine the secondary voltage V_2, the primary current I_1, and the power output P_2.

27. A voltmeter reads 80.0 V when it is connected across the terminals of a sinusoidal power source with f = 1000 Hz. Write the equation for the instantaneous voltage provided by the source. *Ans.* $v = (113\text{ V}) \sin 2000\pi t$ for t in seconds

28. An ac current in a 10 Ω resistance produces thermal energy at the rate of 360 W. Determine the effective values of the current and voltage. *Ans.* 6.0 A, 60 V

29. A 40.0-Ω resistor is connected across a 15.0-V variable-frequency electronic oscillator. Find the current through the resistor when the frequency is (*a*) 100 Hz and (*b*) 100 kHz. *Ans.* (*a*) 0.375 A; (*b*) 0.375 A

30. Solve Problem 16 if the 40.0-Ω resistor is replaced by a 2.00-mH inductor. *Ans.* (*a*) 11.9 A; (*b*) 11.9 mA

31. Solve Problem 16 if the 40.0-Ω resistor is replaced by 0.300-μF capacitor. *Ans.* (*a*) 2.83 mA; (*b*) 2.83 A

32. A coil has resistance 20 Ω and inductance 0.35 H. Compute its reactance and its impedance to an alternating current of 25 cycles/s. *Ans.* 55 Ω, 59 Ω

33. A current of 30 mA is taken by a 4.0-μF capacitor connected across an alternating current line having a frequency of 500 Hz. Compute the reactance of the capacitor and the voltage across the capacitor. *Ans.* 80 Ω, 2.4 V

34. A coil has an inductance of 0.100 H and a resistance of 12.0 Ω. It is connected to a 110-V, 60.0-Hz line. Determine (*a*) the reactance of the coil, (*b*) the impedance of the coil, (*c*) the current through the coil, (*d*) the phase angle between current and supply voltage, (*e*) the power factor of the circuit, and (*f*) the reading of a wattmeter connected in the circuit. *Ans.* (*a*) 37.7 Ω; (*b*) 39.6 Ω; (*c*) 2.78 A; (*d*) voltage leads by 72.3°; (a) 0.303; (*f*) 92.6 W

35. A 10.0-μF capacitor is in series with a 40.0-Λ resistance, and the combination is connected to a 110-V, 60.0-Hz line. Calculate (*a*) the capacitive reactance, (*b*) the impedance of the circuit, (*c*) the current in the circuit, (*d*) the phase angle between current and supply voltage, and (*e*) the power factor for the circuit. *Ans.* (*a*) 266 Ω; (*b*) 269 Λ; (*c*) 0.409 A; (*d*) voltage lags by 81.4°; (*e*) 0.149

36. A circuit having a resistance, an inductance, and a capacitance in series is connected to a 110-V ac line. For the circuit, R = 9.0 Ω, X_L = 28 Ω, and X_C =16 Ω. Compute (*a*) the impedance of the circuit, (*b*) the current, (*c*) the phase angle between the current and the supply voltage, and (*d*) the power factor of the cir-cuit. *Ans.* (*a*) 15 Ω; (*b*) 7.3 A; (*c*) voltage leads by 53°; (*d*) 0.60

37. An experimenter has a coil of inductance 3.0 mH and wishes to construct a circuit whose resonant frequency is 1.0 MHz. What should be the value of the capacitor used? *Ans.* 8.4 pF

38. A circuit has a resistance of 11 Ω, a coil of inductive reactance 120 Ω, and a capacitor with a 120-Ω reactance, all connected in series with a 110-V, 60-Hz power source. What is the potential difference across each circuit element? *Ans.* $V_R = 0.11$ kV, $V_L = V_C = 1.2$ kV

39. A 120-V, 60-Hz power source is connected across an 800-Ω noninductive resistance and an unknown capacitance in series. The voltage drop across the resistor is 102 V. (*a*) What is the voltage drop across the capacitor? (*b*) What is the reactance of the capacitor? *Ans.* (*a*) 63 V; (*b*) 0.50 kΩ

40. A coil of negligible resistance is connected in series with a 90-Ω resistor across a 120-V, 60-Hz line. A voltmeter reads 36 V across the resistance. Find the voltage across the coil and the inductance of the coil. *Ans.* 0.11 kV, 0.76 H

41. Consider an AC voltage source that produces a voltage given by $V = V_{max} \sin(2\pi ft)$, with $V_{max} = 18$ V and $f = 6.2$ Hz. What is the instantaneous value of the voltage at $t = 1.4$ s?

42. The LCR circuit in Figure P22.48 is driven by a voltage source with $V_{AC} = V_{max} \sin(2\pi ft)$, where $V_{max} = 12$ V, $R = 150\ \Omega$, and $C = 25$ nF. (a) If the resonant frequency is $f_{res} = 95$ MHz, what is L? (b) What is the amplitude (the peak value) of the current when $f = f_{res}$?

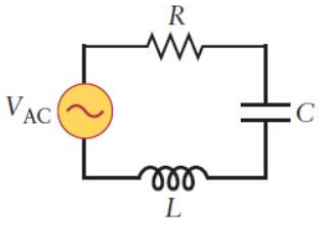


Figure P22.48

43. A step-down transformer is used on a 2.2-kV line to deliver 110 V. How many turns are on the primary winding if the secondary has 25 turns? *Ans.* 5.0×10^2

44. A step-down transformer is used on a 1650-V line to deliver 45 A at 110 V. What current is drawn from the line? Assume 100 percent efficiency. *Ans.* 3.0 A

45. A step-up transformer operates on a 110-V line and supplies a load with 2.0 A. The ratio of the primary and secondary windings is 1:25. Determine the secondary voltage, the primary current, and the power output. Assume a resistive load and 100 percent efficiency. *Ans.* 2.8 kV, 50 A, 5.5 kW

46. Transformers are not limited to two sets of windings. The relationship for winding ratios holds true for multiple sets of coils as well. The AC voltage on the primary coil in Figure P22.77 is $V_P = 120$ V, and the two outputs supply $V_{S1} = 220$ V and $V_{S2} = 12$ V, respectively. If the 220-V secondary coil has 880 turns, what are the number of turns in the 12-V secondary coil and the primary coil?

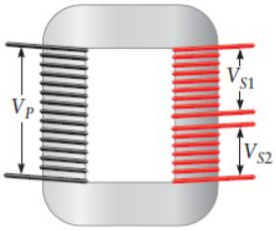


Figure P22.77 A transformer with two secondary coils.